

Lion Guardians Patrols Effort

Technical Guide

Patrol Effort Analysis — Methodology & Calculation Reference
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1. Overview

The **LG Patrols Effort** workflow analyses ranger patrol activity in the Amboseli ecosystem for the **Lion Guardians** programme. It ingests patrol tracks and patrol events from **EarthRanger**, computes patrol effort maps and summary statistics per guardian and patrol type, and delivers an interactive dashboard plus a print-ready Word report.

The workflow produces three maps per group (Patrol Events, Patrol Trajectories, Linear Time Density), five scalar metrics, two charts (bar chart and pie chart), and six summary CSV tables.

Note: All per-group outputs are produced by iterating over a user-chosen grouper: patrol type, patrol serial number, or patrol subject.

2. Dependencies & Prerequisites

2.1 EarthRanger Connection

All patrol data is fetched from an **EarthRanger** instance via `set_er_connection`. The workflow uses `set_patrols_and_patrol_events_params` to configure the query, then calls `get_patrols_from_combined_params` and `get_patrol_observations_from_patrols_df_and_combined_params` to retrieve patrol tracks, and `unpack_events_from_patrols_df_and_combined_params` for events. The sub-page size is 200 records; patrols are not filtered to overlap the date range only.

2.2 Groupers

Three grouper fields are available via `set_groupers`:

Grouper field	EarthRanger source	Typical use
patrol_type	patrol.patrol_type.value	One output per patrol type
patrol_serial_number	patrol.serial_number	One output per patrol serial
patrol_subject	patrol.leader / members	One output per guardian ranger

2.3 Static Geodata Files

Two boundary datasets are downloaded from Dropbox and cached locally:

Dataset	File	Purpose
Group Ranch Boundaries	lg_group_ranch_boundaries.gpkg	Community ranch polygons in Amboseli
Conflict Hotspot Areas	lg_conflict_hotspots.gpkg	Known human–lion conflict hotspot features

Both files use `overwrite_existing: false` (3 retries). After loading, each is reprojected to **EPSG:4326** and annotated with its geometry type before layer creation.

2.4 Word Document Templates

Template file	Purpose
patrol_guardians_cover_page.docx	Report cover page — period, preparer
custom_patrol_template.docx	Per-grouper section — maps, charts, and summary tables

2.5 Base Map Tile Layers

Layer	Opacity	Max zoom
ArcGIS World Hillshade	100 %	20
ArcGIS World Street Map	15 %	20

The hillshade provides full-opacity terrain context. The street map is overlaid at 15 % to show roads and settlement names without obscuring terrain or patrol data.

3. Data Ingestion Pipeline

3.1 Patrol Observations → Relocations → Trajectories

`process_relocations` converts raw patrol observations to a standardised `GeoDataFrame`. Retained columns include patrol identifiers (`patrol_id`, `patrol_start_time`, `patrol_end_time`, `patrol_serial_number`, `patrol_subject`), fix quality (`junk_status`), and point geometry. Three null-island coordinate pairs are filtered out:

- (180.0, 90.0) — boundary sentinel
- (0.0, 0.0) — null-island artefact
- (1.0, 1.0) — common default / test value

A daytime filter (`filter_daytime_patrols`) is applied before trajectory construction, retaining only fixes between 06:00 and 19:00 local time to exclude night-time GPS drift. `relocations_to_trajectory` then connects consecutive fixes per patrol into `LineString` segments, adding `dist_meters`, `speed_kmhr`, `segment_start`, and `segment_end`. The following segment filter is applied:

Filter parameter	Default	Description
<code>min_length_meters</code>	10	Discard segments shorter than 10 m
<code>max_length_meters</code>	100 000	Discard segments longer than 100 km
<code>min_time_secs</code>	10	Discard segments shorter than 10 s
<code>max_time_secs</code>	21 600	Discard segments longer than 6 hours
<code>min_speed_kmhr</code>	1	Discard segments below 1 km/h average speed
<code>max_speed_kmhr</code>	7	Discard segments above 7 km/h average speed

Trajectories are persisted as `trajectories.geoparquet`.

3.2 Patrol Events

`unpack_events_from_patrols_df_and_combined_params` extracts associated events for each patrol. `get_event_type_display_names_from_events` enriches the events `GeoDataFrame` with human-readable event type names (`append_category_names`: duplicates). Events are converted to the user timezone and persisted as `events.geoparquet`.

3.3 Temporal Index & Column Renaming

`add_temporal_index` keys the trajectory `GeoDataFrame` to `extra_patrol_start_time`, grouped by the configured groupers, enabling per-group iteration. Four columns are then renamed via `map_columns` (`raise_if_not_found`: true):

Original column	Renamed to
extra__patrol_type__value	patrol_type
extra__patrol_serial_number	patrol_serial_number
extra__patrol_status	patrol_status
extra__patrol_subject	patrol_subject

The renamed GeoDataFrame is split into per-group partitions by `split_groups`. All downstream map and metric tasks iterate over these partitions via `mapvalues`.

3.4 Trajectory Colormap

A user-selectable **Trajectory Category** field determines how patrol tracks are coloured. Available options are `patrol_type`, `patrol_subject`, and `patrol_serial_number`. `apply_color_map` maps the chosen column to the **Paired** colormap, writing the result to `patrol_traj_colormap`. Similarly, patrol events are coloured by `event_type` using the **Accent** colormap (`event_type_colormap`).

4. Static Map Layers

Three static layers are built once and composited onto every group-level map to provide spatial context.

4.1 Layer Styles

Layer	Colour (RGB)	Opacity	Filled	Notes
Group Ranch Boundaries	(169, 169, 169) grey	45 %	No	Outline only, line width 1.25
Conflict Hotspots	(220, 20, 60) crimson	45 %	Yes	Point radius 2.05, line width 1.25
Hotspot Text Labels	(20, 20, 20) near-black	—	—	Arial, 1 000 m base, 40–75 px clamp, centroid-anchored

5. Map Outputs — Methodology

5.1 Patrol Events Map

`create_scatterplot_layer` renders each patrol event as a point marker, coloured by `event_type_colormap` (Accent colormap). Point radius is 5 px at 75 % opacity with stroked outlines. Before rendering, geometric outliers are removed via `exclude_geom_outliers` (z-threshold: 3) and null geometries are dropped. The event layer is combined with the three static boundarylayers. The map is auto-zoomed to the event extent (expansion factor 1.05) and persisted as HTML (suffix: `events`), then converted to PNG at 2× scale with a 40 s tile-load wait.

5.2 Patrol Trajectories Map

`create_path_layer` renders patrol track segments coloured by the user-selected Trajectory Category column:

Property	Value
Colour	<code>patrol_traj_colormap</code> (Paired colormap, per-category)
Width	1.85 px, min 2 px, max 6 px (screen-space pixels)

Property	Value
Cap / Join	Rounded
Opacity	55 %

The path layer is combined with the three static layers and auto-zoomed to the trajectory extent (expansion factor 1.05). The map is persisted as HTML (suffix: patrol_trajectories). PNG screenshot generation is available but disabled by default in the current workflow version.

5.3 Linear Time Density Map

create_meshgrid builds a raster grid over the full patrol trajectory extent. calculate_linear_time_density then computes the time-weighted density of patrol coverage. Parameters:

Parameter	Value	Meaning
percentiles	50, 60, 70, 80, 90, 100	Contour probability thresholds extracted as polygons
intersecting_only	false	Meshgrid covers the full AOI extent

NaN-percentile rows are dropped and the result is sorted ascending. Contour polygons are coloured with the **RdYlGn** diverging colormap (innermost 50th percentile = red, outermost 100th = green) at 45 % opacity. The map is persisted as HTML (suffix: time_density) and converted to PNG at 2× scale with a 40 s tile-load wait.

6. Summary Metrics

6.1 Per-Guardian Statistics

summarize_df aggregates trajectory data grouped by patrol_subject:

Output column	Source column	Aggregator	Output unit
no_of_patrols	extra__patrol_id	nunique	count
total_distance	dist_meters	sum	km
total_time	timespan_seconds	sum	h

A separate summary counts the number of events per guardian (id nunique). Results are persisted as CSVs. A pivot table is also generated — events pivoted by event_type_display — to show each guardian's event breakdown.

6.2 Patrol Type Summary

The same three aggregations (no_of_patrols, total_distance, total_time) are computed grouped by patrol_type. This shows comparative effort across different patrol activity categories.

6.3 Event Type Summary

Events are summarised by event_type_display using id nunique, giving a count of distinct events per type across the analysis period.

6.4 Monthly Summary

`extract_date_parts` extracts month, day, and month_name from `extra__patrol_start_time`. Patrol effort is then summarised by month_name (`no_of_patrols`, `total_distance`, `total_time`) to reveal seasonal patrol patterns.

6.5 Scalar Dashboard Widgets

Five scalar metrics are computed per group from the trajectory data and displayed as single-value widgets (1 decimal place):

Widget title	Source column	Aggregator	Unit
Total Patrols	<code>extra__patrol_id</code>	<code>nunique</code>	count
Total Time	<code>timespan_seconds</code>	<code>sum</code>	h
Total Distance	<code>dist_meters</code>	<code>sum</code>	km
Average Speed	<code>speed_kmhr</code>	<code>mean</code>	km/h
Max Speed	<code>speed_kmhr</code>	<code>max</code>	km/h

7. Event Charts

7.1 Time Series Bar Chart

`draw_time_series_bar_chart` plots patrol event counts over time. X-axis: time (event timestamp); Y-axis: `event_type_display`; aggregation: `count`; category colour: `event_type_colormap` (Accent colormap). The chart is persisted as HTML (suffix: `patrol_events_time_series_bar_chart`) and converted to PNG.

7.2 Pie Chart

`draw_pie_chart` shows the proportional breakdown of events by `event_type_display`, using `event_type_colormap` for slice colours and displaying raw counts as text labels. The chart is persisted as HTML (suffix: `patrols_pie_chart`) and converted to PNG.

8. Word Report (.docx)

8.1 Cover Page

`create_guardians_ctx_cover` builds the cover context (report period, *Ecoscope* as preparer). `create_context_page_lg` renders it into `context_page.docx` using the `patrol_guardians_cover_page.docx` template.

8.2 Per-Grouper Sections

`guardians_ctx` assembles a context dict per group containing maps (events map, trajectories map, time density map), charts (pie chart, bar chart), and CSV data (monthly efforts, guardian event pivot, guardian events, guardian patrol stats, event type efforts, trajectory DataFrame). `generate_guardians_report` renders each section from the `custom_patrol_template.docx` template. Image boxes: **3.85 × 6.48 inches**. `validate_images: true` catches missing PNGs before rendering.

8.3 Document Merge

merge_cl_files concatenates the cover page and all per-grouper sections into a single Word file saved to the results directory.

9. Interactive Dashboard

gather_dashboard assembles the patrol dashboard from ten widget groups:

Widget	Type	Source task
Total Distance	Single value	total_patrol_dist → patrol_dist_grouped_widget
Average Speed	Single value	avg_speed → avg_speed_grouped_widget
Max Speed	Single value	max_speed → max_speed_grouped_widget
Total Patrols	Single value	total_patrols → total_patrols_grouped_sv_widget
Total Time	Single value	total_patrol_time → patrol_time_grouped_widget
Patrol Events Map	Map	draw_events → events_grouped_map_widget
Trajectories Map	Map	trajs_ecomap → trajs_grouped_map_widget
Events Bar Chart	Plot	patrol_events_bar_chart → grouped_bar_plot_widget_merge
Events Pie Chart	Plot	patrol_events_pie_chart → patrol_events_pie_widget_grouped
Time Density Map	Map	td_ecomap → td_grouped_map_widget

Note: Widget tasks use skipif: [never] so the dashboard always assembles, even when some groups have no data.

10. Output Files

All files are written to \$ECOSCOPE_WORKFLOWS_RESULTS.

File / pattern	Format	Content
events.geoparquet	GeoParquet	Patrol events with display names and timezone-converted timestamps
trajectories.geoparquet	GeoParquet	Patrol segments with speed_kmhr, dist_meters
_events.html	HTML	Interactive patrol events map
_patrol_trajectories.html	HTML	Interactive patrol trajectories map
_time_density.html	HTML	Interactive linear time density map
_events.png	PNG	2× screenshot of patrol events map
_time_density.png	PNG	2× screenshot of time density map
_patrols_pie_chart.png	PNG	2× screenshot of event type pie chart
_bar_chart.png	PNG	2× screenshot of events time series bar chart
_guardian_patrol.csv	CSV	Per-guardian patrol effort (patrols, distance, time)

File / pattern	Format	Content
_guardian_events.csv	CSV	Per-guardian event count
_pivot_guardian_events.csv	CSV	Guardian × event type pivot table
_event_types.csv	CSV	Per-event-type count
_monthly_patrol_efforts.csv	CSV	Monthly patrol effort summary
context_page.docx	Word	Rendered report cover page
.docx	Word	Per-grouper report section
.docx	Word	Final combined Word report

11. Workflow Execution Logic

11.1 Skip Conditions

Two default skip conditions apply to every task (task-instance-defaults):

- **any_is_empty_df** — skips the task (and all dependants) when any input DataFrame is empty, handling patrol types or periods with no data gracefully.
- **any_dependency_skipped** — propagates skips downstream automatically.

Widget and map-widget tasks override this with skipif: [never] to ensure the dashboard always assembles.

11.2 Data Flow Summary

Stage	Tasks
Setup	ER connection, time range, timezone, groupers, base maps
Geodata download	2 boundary files + 2 Word templates from Dropbox
Static layers	Ranch, hotspot, hotspot text layers
Patrol ingest	Params → prefetch → observations → events → rename → convert TZ
Trajectories	Relocations → trajectories → temporal index → rename → split groups
Events branch	Filter → temporal index → colormap → rename → outlier removal → scatter layer → map → HTML → PNG → widget
Trajectories branch	Speed format → rename → colormap → path layer → map → HTML → widget
Time density branch	Meshgrid → LTD → drop NaN → sort → colormap → GeoJSON layer → map → HTML → PNG → widget
Metrics branch	5 scalar widgets; 5 CSV summaries (guardian patrol, guardian events, event type, monthly, pivot)
Charts branch	Time series bar chart + pie chart → HTML → PNG → widgets
Report assembly	Cover page + per-group sections → merge docx
Dashboard	gather_dashboard combines all 10 widgets

12. Software Versions

Package	Version	Role
ecoscope-workflows-core	0.22.17.*	Core task library and workflow engine
ecoscope-workflows-ext-ecoscope	0.22.17.*	Spatial analysis tasks (LTD, relocations, trajectories)
ecoscope-workflows-ext-custom	0.0.40.*	Utility tasks (column mapping, screenshots, pivots)
ecoscope-workflows-ext-ste	0.0.18.*	Summary table and aggregation tasks
ecoscope-workflows-ext-mnc	0.0.7.*	Pivot table task (pivot_df)
ecoscope-workflows-ext-icf	0.0.0.*	ICF domain tasks
ecoscope-workflows-ext-big-life	0.0.8.*	Big Life Foundation domain tasks
ecoscope-workflows-ext-lion-guardians	0.0.6.*	Lion Guardians Word report rendering tasks

Packages are distributed via the prefix.dev conda channel and pinned to patch-compatible versions (*. suffix). The runtime environment is managed by **pixi**.